

steps:-

i) Remove ϵ product

- $C \rightarrow \lambda \Rightarrow C$ is nullable
- $D \rightarrow \lambda \Rightarrow D$ is nullable

For $S \rightarrow ABCBCDA$

$S \rightarrow ABCBCDA \mid ABBCA \mid ABCBA \mid ABBCBA \mid ABCBCA$

- $A \rightarrow CD$
- Remove $C \rightarrow A \rightarrow D$
- Remove $D \rightarrow A \rightarrow C$
- Remove both $\rightarrow \epsilon$

So:- $A \rightarrow CD \mid C \mid D$

$B \rightarrow cb$

Remove $C \rightarrow B \rightarrow b$

$B \rightarrow cb \mid b$

• $D \rightarrow bD \mid \lambda$

$D \rightarrow bD \mid b$

• $S \rightarrow$ all combinations of $ABCBCDA$ without C, D

$A \rightarrow CD \mid C \mid D$

$B \rightarrow cb \mid b$

$C \rightarrow a$

$D \rightarrow bD \mid b$

Step 2:- Remove unit productions

$S \rightarrow \dots$

$A \rightarrow CD \mid a \mid bD \mid b$

$B \rightarrow cb \mid b$

$C \rightarrow a$

$D \rightarrow bD \mid b$